STATS 200A Lecture 5

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$$X : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$\begin{aligned} \mathbb{P}_X(\mathcal{B}) &= \mathbb{P}(X \in \mathcal{B}) \\ &= \mathbb{P}(\{\Omega : X(\omega) \in \mathcal{B}\}) \\ \mathbb{P}((-\infty, x]) &= \mathbb{P}(X \leq 2) \end{aligned}$$

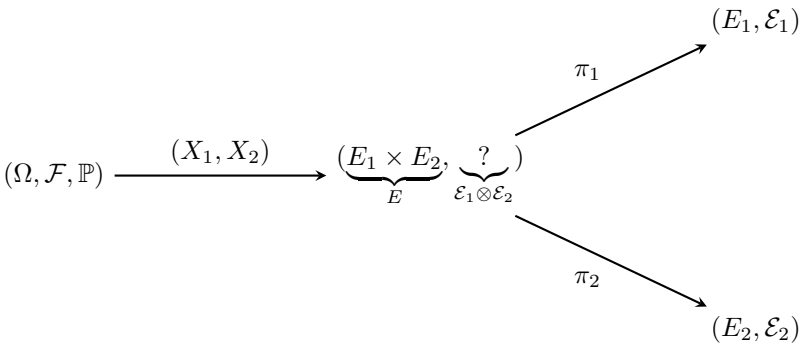
Theorem 5.1 (Quantile-transform/inverse-transform sampling)
For any CDF F, define its *generalized inverse* $F^{-1}(u) = \inf\{x : F(x) \geq u\}$, $0 < u < 1$. If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has CDF F.

$$\mu(A) = \frac{\lambda(A \cap D)}{\lambda(D)}$$

$$X_i : (\Omega, \mathcal{F}, \mathbb{P}) \to (E_i, \mathcal{E}_i)$$

$$X_1 : \Omega \to (E_1, \mathcal{E}_1),\ X_2 : \Omega \to (E_2, \mathcal{E}_2)$$

$$E_1 \times E_2 : \{(x_1, x_2) : x_1 \in E_1, x_2 \in E_2\}$$



$$\mathcal{E}_1 \otimes \mathcal{E}_2 = \sigma(\text{coordinate strips}) = \sigma(\text{measurable rectangles})$$

Rectangles form a π-system

Definition 5.2 (Product σ-algebra)

$$\bigotimes_{i \in I} \mathcal{E}_i := \sigma\left(\left\{\pi_i^{-1}(B) : i \in I, B \in \mathcal{E}_i\right\}\right)$$

This is the initial σ-algebra for the coordinate maps.

Definition 5.3 (Cylinder σ-algebra)

$$\mathcal{S}_T = \sigma\left(\left\{e_t^{-1}(B) : t \in T, B \in \mathcal{B}(\mathbb{R})\right\}\right) = \mathcal{B}(\mathbb{R})^{\bigotimes T}$$

Definition 5.4 (Cylinder set)

$$C = \{f : f(t_1) \in B_1, \ldots, f(t_k) \in B_k\}$$